

- 3 A spring is fixed at one end and attached to the frame of a pulley at the other end. A cable is passed around the wheel of the pulley. The spring is stretched to a fixed length using the cable and pulley.

Fig. 3.1 shows the view from above of the spring, cable and pulley.

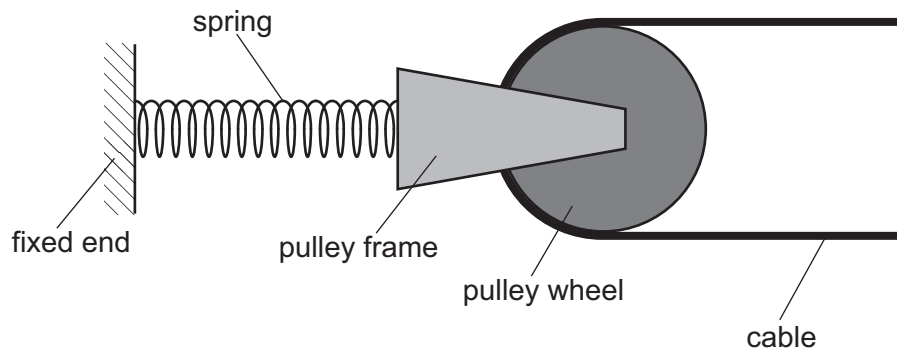


Fig. 3.1

The spring obeys Hooke's law and has a spring constant k of 250 N m^{-1} . A force F acts on the spring. The tension in the cable is T . The pulley is in equilibrium.

- (a) On Fig. 3.2, draw labelled arrows to show the directions of the forces acting on the pulley.

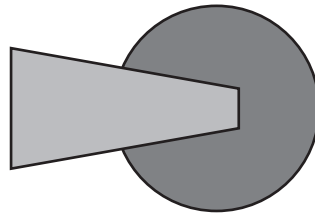


Fig. 3.2

[2]

- (b) The force F is 110 N .

- (i) Determine T .

$T = \dots\dots\dots \text{ N}$ [1]

